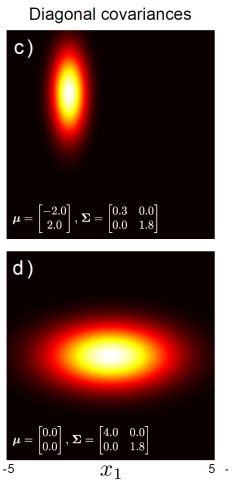
Agenda

$$\Sigma_{\text{diag}} = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}$$

- Earlier
  - Goodness-of-fit when probabilities are not fully specified
  - Multivariate Normal
- Today
  - Meta content
  - Goodness of fit for exponential distribution
  - A word on PCA and t-SNE?
  - Epilogue ??



1

Review

$$\begin{array}{cccccccccc} 8 & 0 & 0 & 1 & 3 & 4 & 0 & 2 & 12 & 5 \\ 1 & 8 & 0 & 2 & 0 & 1 & 9 & 3 & 4 & 5 \\ 3 & 3 & 4 & 7 & 4 & 0 & 1 & 2 & 1 & 2 \end{array}$$

- Can GoF: (Table of counts idea) work with *continuous* distribution?
  - Unlike discrete tables such as Poisson, there are no natural categories
- To motivate this:
  - We know how many accidents on a weekly basis but not when
  - If accidents occur at a constant rate (Poisson), what is the "waiting" time between successive accidents?
    - Accidents at a constant average rate of λ accidents per day (earlier per week)
    - Take the unit to be one day
    - N(t) = the number of accidents in a time window of length t days
    - T = the waiting time (in days) from a starting moment until the next accident

2

Exponential Distribution

- Waiting time, *T* random variable. Distribution == ?
- For the gap between accidents, start the clock at an accident
  - Same Binomial argument as before but with window of length t
  - $P(N(t) = k) = e^{-\lambda t}(\lambda t)^k/k!$
  - Therefore, the probability that the first accident occurred after time *u* is probability that there was no accident in the time interval *(0,u)*
    - $P(T > u) = P(N(u) = 0) = e^{-\lambda u}(\lambda u)^0/0! = e^{(-\lambda u)}$
    - $P(T \leq u) = 1 - P(T > u) = 1 - e^{-\lambda u}$
- This is the exponential distribution with **rate** λ with pdf $f_T(u) = \lambda e^{-\lambda u}, u \geq 0$
  - Continuous analog of waiting for the first "success" (geometric distribution)

3

$$P(T > s + u \mid T > s) = \frac{P(T > s + u)}{P(T > s)} = \frac{e^{-\lambda(s+u)}}{e^{-\lambda s}} = e^{-\lambda u} = P(T > u)$$

Table of counts (continuous)

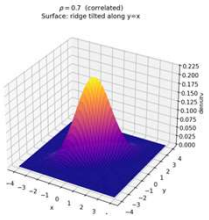
- 210 days of data. We know how many accidents on a weekly basis
  - We are now given finer record (timestamp of accidents): n = number of gaps 90, rate λ= 90/210 = 0.4286 (i.e., average "wait time" of 2.33 days)
  - Null H_0: Is T exponential?? Divide into 4 bins. Bin probabilities: $P(a < \mathrm{T} <= b) = e^{-\lambda a} - e^{-\lambda b}$, $x_0^2 \approx 1.64$, 2 df, $\mathrm{p-value} \approx 0.44$. Do not reject.
- Memoryless property: If you have already waited s days with no accident, the chance of waiting at least u more days is the same as if you had just started waiting

We chop the real line into intervals for table of counts

Gap (days)	0–1	1–2	2–4	≥ 4
Observed O_i	29	25	19	17
Expected $E_i = 90\,p_i$	31.37	20.44	21.99	16.21
$(O_i - E_i)^2/E_i$	0.18	1.02	0.41	0.04

4

Review: Multivariate Normal



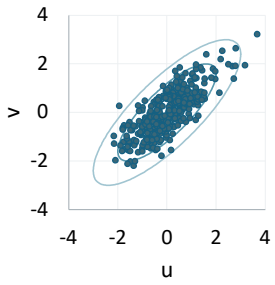
- Equivalent definitions
  1. Joint density of **X** as defined
     - The mean and covariance of random vector is $\boldsymbol{\mu}$ and $\mathbf{C}$
  2. Every linear combination of elements of **X** is normally distributed
  3. **X = AZ + b** for some i.i.d standard normal, an invertible A and a column vector b
- Nomenclature: The (scalar) random variables in the stacked vector are said to be *jointly normal*

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{\frac{d}{2}}\sqrt{\det \mathbf{C}}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\top}\mathbf{C}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)$$

5

Caution: Independence not given

- Consider $Z_1, Z_2 \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$
  - $U = Z_1 \quad V = 0.8Z_1 + 0.6Z_2$
  - This is a linear combination and hence U is jointly normal



$$\mathbf{C} = \begin{bmatrix} 1 & 0.8 \\ 0.8 & 1 \end{bmatrix}$$

6

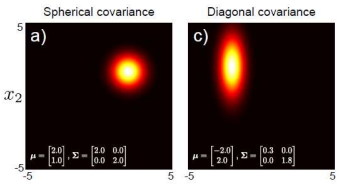
Diagonal Covariance = Independence

$$\Pr(x_1, x_2) = \frac{1}{2\pi\sqrt{|\boldsymbol{\Sigma}|}} \exp\left[-\frac{1}{2}\begin{pmatrix} x_1 & x_2 \end{pmatrix} \boldsymbol{\Sigma}^{-1} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right]$$

$$\Pr(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2} \exp\left[-\frac{1}{2}\begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} \sigma_1^{-2} & 0 \\ 0 & \sigma_2^{-2} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right]$$

$$\Pr(x_1, x_2) = \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left[-\frac{x_1^2}{2\sigma_1^2}\right] \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp\left[-\frac{x_2^2}{2\sigma_2^2}\right]$$

$$\Pr(x_1, x_2) = \Pr(x_1)\Pr(x_2)$$



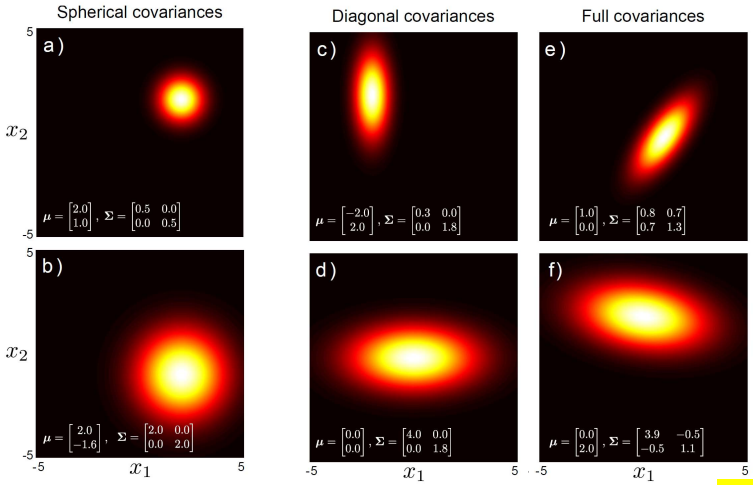
$$\boldsymbol{\Sigma}_{\text{diag}} = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}$$

7

Review

$$\boldsymbol{\Sigma}_{\text{diag}} = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}$$

$$\boldsymbol{\Sigma}_{\text{spher}} = \begin{bmatrix} \sigma^2 & 0 \\ 0 & \sigma^2 \end{bmatrix}$$



$$\boldsymbol{\Sigma}_{\text{full}} = \begin{bmatrix} \sigma_{11}^2 & \sigma_{12}^2 \\ \sigma_{21}^2 & \sigma_{22}^2 \end{bmatrix}$$

We will assume joint normal

8

Review: Marginal Distributions

We will assume joint normal

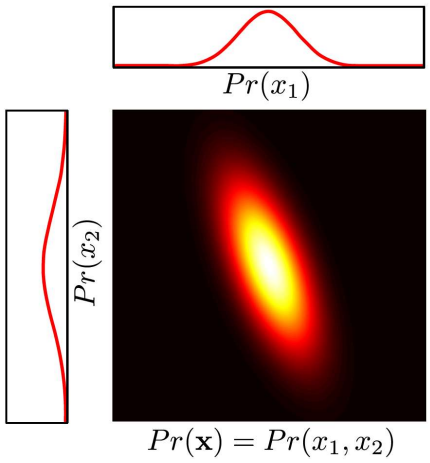
- Marginal distributions of a multivariate normal are also normal

$$\Pr(\mathbf{x}) = \Pr\left(\begin{bmatrix}\mathbf{x}_1\\ \mathbf{x}_2\end{bmatrix}\right)$$

$$\Pr(\mathbf{x}) = \text{Norm}_{\mathbf{x}}\left[\begin{bmatrix}\boldsymbol{\mu}_1\\ \boldsymbol{\mu}_2\end{bmatrix}, \begin{bmatrix}\boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{21}^T\\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22}\end{bmatrix}\right]$$

$$\Pr(\mathbf{x}_1) = \text{Norm}_{\mathbf{x}_1}[\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11}]$$

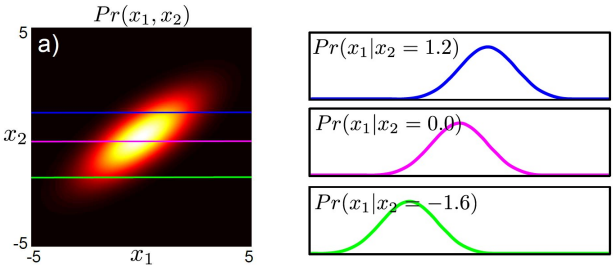
$$\Pr(\mathbf{x}_2) = \text{Norm}_{\mathbf{x}_2}[\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{22}]$$



9

Conditional Distributions

- Conditional distributions of a multivariate normal are also normal



$$\Pr(\mathbf{x}) = \Pr\left(\begin{bmatrix}\mathbf{x}_1\\ \mathbf{x}_2\end{bmatrix}\right) = \text{Norm}_{\mathbf{x}}\left[\begin{bmatrix}\boldsymbol{\mu}_1\\ \boldsymbol{\mu}_2\end{bmatrix}, \begin{bmatrix}\boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12}^T\\ \boldsymbol{\Sigma}_{12} & \boldsymbol{\Sigma}_{22}\end{bmatrix}\right]$$

We will assume joint normal

$$\Pr(\mathbf{x}_1 \mid \mathbf{x}_2) = \text{Norm}_{\mathbf{x}_1}\left[\boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}^T\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2), \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}^T\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{12}\right]$$

$$\Pr(\mathbf{x}_2 \mid \mathbf{x}_1) = \text{Norm}_{\mathbf{x}_2}\left[\boldsymbol{\mu}_2 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{11}^{-1}(\mathbf{x}_1 - \boldsymbol{\mu}_1), \boldsymbol{\Sigma}_{22} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}^T\right]$$

10

Visualization: Projecting High Dimensional Data

- When a multivariate dataset has a large number of variables, analysis and interpretation of the data may be hard.
  - Too many variable pairs, so pairwise correlation may be hard to grasp.
- For convenient visualization and interpretation
  - Reduce the number of variables.
- How to reduce number of variables while capturing most of the information in the data
  - Information (variance) PCA, Factor analysis, Random Projection, t-SNE

11

T-distributed stochastic neighbourhood embedding

- Another data projection method, specifically designed for visualizing high dimensional data in two dimensions.
- Uses a probability distribution in latent space that has heavier tails
  - Eliminates the unwanted attractive forces between points that are relatively far in the high dimensional space.
  - Preserves local similarities and clusters better than PCA
  - Creates non-linear projection
  - Reference: Kevin Murphy Section 20.4.10 Probabilistic ML book

13